



Researcher
HITL

Propose ideas
e.g., cyclic shifts

Interpret discoveries
generalize and prove

Revise the task
between campaigns

task specification

inspect witnesses

01 TARGET

n	$d_\ell - d_u$
7	5 - 6
9	7 - 8
11	9 - 10

Reference tables
Known bounds

02 EVOLVE

LLM evolves
a program



Generators

Exact evaluator

feedback

03 VERIFY



SAT + DRAT
proof check

Certify nonexistence

04 DISCOVER



Tighten code bounds

$[[11, 1, 10; 8]]_2$

115

witnesses

10

refutations

nonexistence targets